

CSA0301

DATA STRUCTURES

Activity 2

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1. Student Record Stack:

```
#include <stdio.h>
#include <string.h>

#define MAX 5

struct Student
{
    int id;
    char name[50];
    char dept[50];
};

struct Student stack[MAX];
int top = -1;

// Push Operation
void push()
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }

    top++;

    printf("Enter Student ID: ");
    scanf("%d", &stack[top].id);
```

```

printf("Enter Student Name: ");
scanf(" %[^\\n]", stack[top].name);

printf("Enter Department: ");
scanf(" %[^\\n]", stack[top].dept);

printf("Student record pushed successfully.\\n");
}

// Pop Operation
void pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\\n");
        return;
    }

    printf("\\nDeleted Record:\\n");
    printf("ID: %d\\n", stack[top].id);
    printf("Name: %s\\n", stack[top].name);
    printf("Department: %s\\n", stack[top].dept);

    top--;
}

// Peek Operation
void peek()
{
    if (top == -1)
    {
        printf("Stack is Empty\\n");
        return;
    }

    printf("\\nTop Student Record:\\n");
    printf("ID: %d\\n", stack[top].id);
    printf("Name: %s\\n", stack[top].name);
    printf("Department: %s\\n", stack[top].dept);
}

// Display Operation
void display()

```

```

{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nStudent Records in Stack:\n");

    for (int i = top; i >= 0; i--)
    {
        printf("\nRecord %d\n", top - i + 1);
        printf("ID: %d\n", stack[i].id);
        printf("Name: %s\n", stack[i].name);
        printf("Department: %s\n", stack[i].dept);
    }
}

// Main Function
int main()
{
    int choice;

    while (1)
    {
        printf("\n----- Student Stack Menu ----- \n");
        printf("1. Push Student\n");
        printf("2. Pop Student\n");
        printf("3. Peek Student\n");
        printf("4. Display Students\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                push();
                break;

            case 2:
                pop();
                break;

```

```

        case 3:
            peek();
            break;

        case 4:
            display();
            break;

        case 5:
            return 0;

        default:
            printf("Invalid Choice\n");
    }
}

return 0;
}

```

2. Employee Record Stack:

```

#include <stdio.h>

#define MAX 5

struct Employee
{
    int id;
    char name[50];
    char gender[10];
    char job[50];
};

struct Employee stack[MAX];
int top = -1;

// Push Operation
void push()
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
    }
}

```

```

        return;
    }

    top++;

    printf("Enter Employee ID: ");
    scanf("%d", &stack[top].id);

    printf("Enter Employee Name: ");
    scanf(" %[^\\n]", stack[top].name);

    printf("Enter Gender: ");
    scanf(" %[^\\n]", stack[top].gender);

    printf("Enter Job Title: ");
    scanf(" %[^\\n]", stack[top].job);

    printf("Employee record pushed successfully.\\n");
}

```

// Pop Operation

```

void pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\\n");
        return;
    }

    printf("\\nDeleted Employee Record:\\n");
    printf("ID: %d\\n", stack[top].id);
    printf("Name: %s\\n", stack[top].name);
    printf("Gender: %s\\n", stack[top].gender);
    printf("Job Title: %s\\n", stack[top].job);

    top--;
}

```

// Peek Operation

```

void peek()
{
    if (top == -1)
    {

```

```

        printf("Stack is Empty\n");
        return;
    }

    printf("\nTop Employee Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Name: %s\n", stack[top].name);
    printf("Gender: %s\n", stack[top].gender);
    printf("Job Title: %s\n", stack[top].job);
}

// Display Operation
void display()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nEmployee Records:\n");

    for (int i = top; i >= 0; i--)
    {
        printf("\nRecord %d\n", top - i + 1);
        printf("ID: %d\n", stack[i].id);
        printf("Name: %s\n", stack[i].name);
        printf("Gender: %s\n", stack[i].gender);
        printf("Job Title: %s\n", stack[i].job);
    }
}

int main()
{
    int choice;

    while (1)
    {
        printf("\n----- Employee Stack Menu ----- \n");
        printf("1. Push Employee\n");
        printf("2. Pop Employee\n");
        printf("3. Peek Employee\n");
        printf("4. Display Employees\n");
    }
}

```

```

printf("5. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        push();
        break;

    case 2:
        pop();
        break;

    case 3:
        peek();
        break;

    case 4:
        display();
        break;

    case 5:
        return 0;

    default:
        printf("Invalid Choice\n");
}
}

return 0;
}

```

3. Book Record Stack:

```

#include <stdio.h>

#define MAX 5

struct Book
{
    int id;

```

```

    char title[50];
    char author[10];
    int price;
};

Struct Book stack[MAX];
int top = -1;

// Push Operation
void push()
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }

    top++;

    printf("Enter Book ID: ");
    scanf("%d", &stack[top].id);

    printf("Enter Book Title: ");
    scanf(" %s", stack[top].title);

    printf("Enter Author name: ");
    scanf(" %s", stack[top].author);

    printf("Enter Book Price: ");
    scanf("%d", &stack[top].price);

    printf("Book record pushed successfully.\n");
}

// Pop Operation
void pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\n");
        return;
    }
}

```



```

    printf("\nDeleted Book Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Book title: %s\n", stack[top].title);
    printf("Book Author: %s\n", stack[top].author);
    printf("Price: %d\n", stack[top].price);

    top--;
}

// Peek Operation
void peek()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nTop Book Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Book title: %s\n", stack[top].title);
    printf("Book Author: %s\n", stack[top].author);
    printf("Price: %d\n", stack[top].price);
}

// Display Operation
void display()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nBook Records:\n");

    for (int i = top; i >= 0; i--)
    {
        printf("\nRecord %d\n", top - i + 1);
        printf("ID: %d\n", stack[i].id);
        printf("Book Title: %s\n", stack[i].title);
        printf("Book Author: %s\n", stack[i].author);
        printf("Price: %d\n", stack[i].price);
    }
}

```

```

    }
}

int main()
{
    int choice;

    while (1)
    {
        printf("\n----- Book Stack Menu ----- \n");
        printf("1. Push a Book\n");
        printf("2. Pop a Book\n");
        printf("3. Peek a book\n");
        printf("4. Display Books\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                push();
                break;

            case 2:
                pop();
                break;

            case 3:
                peek();
                break;

            case 4:
                display();
                break;

            case 5:
                return 0;

            default:
                printf("Invalid Choice\n");
        }
    }
}

```

```
    }  
    return 0;  
}
```

4. Hospital Patient Stack:

```
#include <stdio.h>
```

```
#define MAX 5
```

```
struct Patient  
{  
    int id;  
    char name[50];  
    int age;  
    char bloodGroup[10];  
};
```

```
struct Patient stack[MAX];  
int top = -1;
```

```
// Push Operation
```

```
void push()  
{  
    if (top == MAX - 1)  
    {  
        printf("Stack Overflow\n");  
        return;  
    }  
}
```

```
top++;
```

```
printf("Enter Patient ID: ");  
scanf("%d", &stack[top].id);
```

```
printf("Enter Patient Name: ");  
scanf(" %[^\\n]", stack[top].name);
```

```
printf("Enter Age: ");  
scanf("%d", &stack[top].age);
```

```
printf("Enter Blood Group: ");  
scanf(" %[^\\n]", stack[top].bloodGroup);
```

```

    printf("Patient record pushed successfully.\n");
}

// Pop Operation
void pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\n");
        return;
    }

    printf("\nDeleted Patient Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Name: %s\n", stack[top].name);
    printf("Age: %d\n", stack[top].age);
    printf("Blood Group: %s\n", stack[top].bloodGroup);

    top--;
}

// Peek Operation
void peek()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nTop Patient Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Name: %s\n", stack[top].name);
    printf("Age: %d\n", stack[top].age);
    printf("Blood Group: %s\n", stack[top].bloodGroup);
}

// Display Operation
void display()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
    }
}

```

```

        return;
    }

    printf("\nPatient Records:\n");

    for (int i = top; i >= 0; i--)
    {
        printf("\nRecord %d\n", top - i + 1);
        printf("ID: %d\n", stack[i].id);
        printf("Name: %s\n", stack[i].name);
        printf("Age: %d\n", stack[i].age);
        printf("Blood Group: %s\n", stack[i].bloodGroup);
    }
}

int main()
{
    int choice;

    while (1)
    {
        printf("\n----- Patient Stack Menu ----- \n");
        printf("1. Push Patient\n");
        printf("2. Pop Patient\n");
        printf("3. Peek Patient\n");
        printf("4. Display Patients\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                push();
                break;

            case 2:
                pop();
                break;

            case 3:
                peek();

```

```

        break;

    case 4:
        display();
        break;

    case 5:
        return 0;

    default:
        printf("Invalid Choice\n");
    }
}

return 0;
}

```

5. Product Inventory Stack:

```

#include <stdio.h>

#define MAX 5

struct Product
{
    int id;
    char name[50];
    char category[50];
    float price;
};

struct Product stack[MAX];
int top = -1;

// Push Operation
void push()
{
    if (top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }
}

```

```

    top++;

    printf("Enter Product ID: ");
    scanf("%d", &stack[top].id);

    printf("Enter Product Name: ");
    scanf(" %[^\\n]", stack[top].name);

    printf("Enter Category: ");
    scanf(" %[^\\n]", stack[top].category);

    printf("Enter Price: ");
    scanf("%f", &stack[top].price);

    printf("Product record pushed successfully.\\n");
}

// Pop Operation
void pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\\n");
        return;
    }

    printf("\\nDeleted Product Record:\\n");
    printf("ID: %d\\n", stack[top].id);
    printf("Name: %s\\n", stack[top].name);
    printf("Category: %s\\n", stack[top].category);
    printf("Price: %.2f\\n", stack[top].price);

    top--;
}

// Peek Operation
void peek()
{
    if (top == -1)
    {
        printf("Stack is Empty\\n");
        return;
    }

```

```

    }

    printf("\nTop Product Record:\n");
    printf("ID: %d\n", stack[top].id);
    printf("Name: %s\n", stack[top].name);
    printf("Category: %s\n", stack[top].category);
    printf("Price: %.2f\n", stack[top].price);
}

// Display Operation
void display()
{
    if (top == -1)
    {
        printf("Stack is Empty\n");
        return;
    }

    printf("\nProduct Records:\n");

    for (int i = top; i >= 0; i--)
    {
        printf("\nRecord %d\n", top - i + 1);
        printf("ID: %d\n", stack[i].id);
        printf("Name: %s\n", stack[i].name);
        printf("Category: %s\n", stack[i].category);
        printf("Price: %.2f\n", stack[i].price);
    }
}

// Main Function
int main()
{
    int choice;

    while (1)
    {
        printf("\n----- Product Stack Menu ----- \n");
        printf("1. Push Product\n");
        printf("2. Pop Product\n");
        printf("3. Peek Product\n");
        printf("4. Display Products\n");
        printf("5. Exit\n");
    }
}

```



```
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        push();
        break;

    case 2:
        pop();
        break;

    case 3:
        peek();
        break;

    case 4:
        display();
        break;

    case 5:
        return 0;

    default:
        printf("Invalid Choice\n");
}
}

return 0;
}
```